

HEXBOARD SEQUENCER MODE QUICK MANUAL

Overview

Sequencer mode is a separate mode from Keyboard mode. It is a single-track, 32-step sequencer. Each step can hold up to 4 tuning-aware notes, its own note length, its own velocity, its own probability, and can play through MIDI or the onboard synth.

Enter And Exit

- From the main menu, choose Sequencer.
- In the Sequencer menu, choose Keyboard to return to normal keyboard mode.

Main Controls

- Steps: the first 8 playable keys of the first 4 rows are steps 1-32.
- Play/Stop: the top-right key starts and stops playback on a short press.
- Play/Stop also deselects the current step if one is selected.
- Performance monitor: hold the top-right Play/Stop key for 2 seconds to show audio engine load, memory, storage, and MIDI receive stats; release it to close the monitor.
- Overview: the 9th key on the second row pages through an 8-step note view.
- Overview stays lit white while idle and becomes brighter white while the overview is open.
- Blue action key: the far-right key on the second row does undo on short press and restores the selected step to the state it had when selected, including notes, length, velocity, probability, and Tie; it clears the selected step on a 1-second hold.
- Tools: the last button on the third row opens a step-tools picker for the selected step.
- Tools stays lit white while idle and becomes brighter white while the tools picker is open.
- Encoder:
 - Turn while a step is selected to change that step's length quickly.
 - Press during file or folder naming to save the name.

Step Editing

- Press a step once to select it.
- Press the same step again to deselect it.
- With a step selected, play note keys on the lower part of the board to add or remove notes from that step.
- The edit screen shows a compact info line for the selected step: L ...% V ... P ...%, then the notes.
- Step notes are kept sorted from lowest to highest.
- In 12 EDO, note labels use normal note names like C3.

- In other tunings, note labels use step.octave style labels like 7.4.
- Tap Preview can make a tapped step audition its stored notes when selected.
- Tied steps do not audition from Tap Preview.

Step Tools Picker

- Press the last button on the third row with a step selected to open tools.
- The tools screen shows the selected step number, a compact L ...% V ... P ...% line, and the current notes.
- The tool buttons move to the lower part of the board so the 4 step rows stay visible at the top while the picker is open.
- Tool selection keys light blue at the sequencer medium light level.
- While the picker is open, tapping another step pad switches the selected step and keeps the picker open.
- Current tool choices are: Len, Vel, Oct+, Oct-, Prob, Tie, Copy
- The tools screen also shows: Cancel/Finished
- Len opens exact length entry for the selected step.
- Oct+ transposes the selected step up 1 tuning cycle and stays in the tools screen.
- Oct- transposes the selected step down 1 tuning cycle and stays in the tools screen.
- Oct+ and Oct- are limited to displayed octave range 0 through 9.
- If Oct+ or Oct- hits that range limit, the range message closes back to tools.
- While that range message is visible, repeated key presses only extend the timeout and do not trigger normal keyboard actions.
- Vel opens exact velocity entry for the selected step.
- Prob opens exact probability entry for the selected step.
- Tie toggles a per-step continuation flag and stays in the tools screen.
- When a step is tied, note-oriented displays show T instead of note labels.
- Tap Preview and manual step preview stay silent on tied steps.
- Copy opens target-step selection and copies the source step's notes, length, velocity, probability, and Tie state into the tapped destination step, then keeps the tools picker open on that destination step.
- Cancel/Finished closes the picker when you are done.

Step Velocity

- Velocity is stored per step.
- Range is 0-127.

- Default step velocity is 96.
- Use Vel from the step tools picker to edit it.
- Velocity entry uses the encoder.
- Turn the encoder to move through: 0, 5, 10, 15, ... 125, 127
- Press the encoder to save.
- Playback, tap preview, save, and load all use the stored step velocity.

Step Probability

- Probability is stored per step.
- Range is 0-100%.
- Default step probability is 100%.
- Use Prob from the step tools picker to edit it.
- Probability entry uses the encoder.
- Turn the encoder to move through: 0, 5, 10, 15, ... 100
- Press the encoder to save.
- Playback, save, and load all use the stored step probability.
- Tap Preview ignores probability and still auditions the step.
- Tied steps are the exception: Tap Preview stays silent on them.

Note Length

- With a step selected, turn the encoder for quick length changes.
- Use Len from the step tools picker to open exact length entry.
- Available lengths are: 0%, 25%, 50%, 75%, 100%, 150%, 200%, 250%, 300%, 350%, 400%, 500%, 600%, 700%, 800%, 900%, 1000%

Exact length screen:

- press number keys to enter the value
- the OLED shows separate Current : and New : values
- digits fill New : without overwriting Current :
- up to 4 digits are allowed
- exact-length keypad keys light blue at the sequencer medium light level
- <- deletes the previous digit and is on the eighth-row first key
- CANCEL exits without saving and uses the same lower cancel area as the tools picker
- press the encoder to save the New : value, or keep the current value if New : is blank

Playback

- Tempo is in Playback Settings and is the internal step clock.
- Each step is a 16th note at the current tempo.
- MIDI Sync is in Playback Settings.
- Clock Source chooses: Internal or External MIDI
- With Clock Source set to External MIDI, incoming MIDI clock drives the step timing and incoming MIDI Start/Stop controls transport.
- Send Clock can send MIDI clock while the sequencer is playing in Internal mode.
- Send Transport can send MIDI Start/Stop when local transport is used in Internal mode.
- Hold Play/Stop for 2 seconds to show the performance monitor while held.
- Steps sets the active playback range from 1 to 32.
- Step lights beyond the active Steps count stay off, even if those steps still contain notes.
- Direction choices are: Forward, Backward, Ping-Pong, Random, Brownian, Drunk
- Play Type chooses: MIDI or OB Synth
- Sequencer note entry, preview, and playback follow the current Keyboard tuning and transpose.
- During transport playback, the lower playable keyboard lights the notes that are currently sounding for their full held duration.
- A tied step does not retrigger a note during transport playback.
- A tied step continues the nearest earlier eligible non-tied step from the same pattern pass when that source note is still active.
- If no valid earlier source is active, the tied step is silent.
- Tie does not wrap from the end of the pattern back to step 1 in v1.

Menu Items

- When a menu page is open, hold the bottom command button below the encoder to temporarily make the top two command buttons act like one encoder step up or down. Releasing that bottom button returns them to normal behavior.
- While editing a value, the top shortcut button increases that value and the lower shortcut button decreases it.
- New: clears to a blank untitled sequence.
- Save: saves to the current file, or opens folder selection and naming if the sequence is untitled.
- Save New: saves as a new file in the selected folder.
- Load: loads a saved sequence.
- Revert: reloads the last saved version of the current sequence.

- Seq Lights: Accent Every, Step Color, Step Hue
- Color Options: opens the shared LED color settings page; its Animation setting also affects the lower keyboard note lights that transport playback triggers
- Playback Settings: Steps, Direction, Tempo, Play Type, MIDI Sync
- Playback Settings > MIDI Sync: Clock Source, Send Clock, Send Transport
- File Management: Rename File, Rename Folder, Create Folder, Delete File, Delete Folder, USB Backup
- Tap Preview: audition step contents when a step is tapped
- Synth Options: opens the shared synth settings page
- Update Firmware: reboots to the firmware update bootloader

Seq Lights

- Accent Every emphasizes every Nth step, starting from step 1.
- Choices are: Off, 2, 3, 4, 5, 6, 7, 8
- Accent Every uses brightness changes only and does not change hue.
- Default Accent Every is 4.
- Step Color choices are: Note, Regular
- Default Step Color is Regular.
- Sequencer step lights use this brightness ladder: off, low, medium, high, highest
- The current step mapping is: empty step without accent = off empty step with accent = medium step with note = medium accented step with note = high selected step with note = high playing step with note = highest
- A tied step uses the same programmed-step brightness behavior as a step with a note.
- During transport playback, the lower playable keyboard lights the currently sounding notes, including ties that keep an earlier note held.
- In Note mode, programmed steps use note-based colors with that brightness ladder.
- In Regular mode, programmed steps use the selected Step Hue with that brightness ladder.
- Empty unselected steps without accent stay off in both modes.
- A selected or playing empty step lights at the highest level so selection and playback are still visible.
- Step Hue is available only in Regular mode.
- Step Hue choices are: Red, Orange, Yellow, Lime, Green, Teal, Cyan, Lt Blue, Blue, Indigo, Purple, Magenta, Pink
- Default Step Hue is Indigo.
- Step Hue previews live while you scroll through its choices.

Saving And Files

- Sequences are stored under /Sequences in onboard flash.
- Sequence files store note pitches as tuning-relative step values instead of raw MIDI note numbers.
- Sequence files also store: Tempo, Steps, Play Type, Direction, per-step notes, per-step length, per-step velocity, per-step probability, and per-step Tie.
- Tap Preview is a general sequencer setting and is not stored per sequence file. It is saved with the board profile and Auto-Save settings system.
- Clock Source, Send Clock, and Send Transport are general sequencer settings and are not stored per sequence file. They are saved with the board profile and Auto-Save settings system.
- Seq Lights settings are general sequencer settings and are not stored per sequence file. They are saved with the board profile and Auto-Save settings system.
- Color Options settings are shared with Keyboard mode and use the same board profile and Auto-Save settings system.
- In those shared settings, Dim Bright affects out-of-scale note LEDs only when Scale Lock is Off.
- On startup, the sequencer restores the last loaded sequence file when that remembered path is still valid.
- File Management > USB Backup starts a dedicated USB serial backup session for the / Sequences tree.
- If you back out of the USB Backup page while a session is active, the sequencer warns that leaving the page closes USB Backup and asks whether you want to continue.
- If you choose Stop Session while a session is active, the sequencer warns that stopping USB Backup ends any current transfer and asks whether you want to continue.
- While USB Backup is active, the hex buttons are temporarily disabled and their LEDs are blanked so you do not accidentally trigger other sequencer actions. The encoder still works for the menu.
- The computer-side tools are: `scripts/hexboard_backup.py` for the command line and `scripts/hexboard_backup_gui.py` for a desktop GUI.
- For skilled computer users, the simplest desktop setup is: install Python 3, install `pyserial`, then launch the GUI.
- The repo root also includes simple double-click launchers: `Launch HexBoard Backup.command` for macOS and `Launch HexBoard Backup.bat` for Windows.
- The backup tools can work on the full /Sequences tree or on a selected file or folder.
- In the desktop GUI, you can multi-select remote items with Shift-click and delete them together.

- The desktop GUI now has four action rows overall: one row beside the serial port dropdown for Refresh Ports and Refresh Remote, then three rows below for the rest of the actions.
- The rows below group: full backup actions, download and upload actions, then rename and delete actions.
- The GUI has separate Rename File and Rename Folder buttons.
- GUI renaming only allows simple sequencer-style names: letters, numbers, spaces, and –.
- GUI renaming rejects periods and limits names to 20 characters.
- If a rename is not allowed, the GUI shows the full rename rules, points out the broken rule(s), and then returns to the rename prompt with your previous entry still filled in.
- GUI Upload Folder always merges into the target folder on the board. It does not delete the existing remote folder first.
- During GUI Upload Folder, any existing remote file conflict is shown one file at a time so you can overwrite, skip that file, or cancel the folder upload.
- GUI Upload File warns before replacing an existing remote file, whether that file was selected directly or is already inside the selected target folder.
- The warning also tells you to select a folder first if you want to copy the file into that folder instead of replacing a selected file.
- When multiple remote items are selected, single-target actions like Download, Upload, and Rename are disabled until only one item is selected again.
- The GUI remembers the last local folder used for Upload File and Upload Folder, and it also remembers the last backup folder used for Restore Backup. Those locations return after reopening the desktop tool.
- The top GUI status line shows active operations, including Refresh Remote.
- The Size Bytes column shows raw file sizes in bytes.
- While that session is active, Save, Revert, and file-browser actions are blocked until the session is stopped.
- File browsers show up to 8 file/folder rows at a time.
- ^ more ^ and v more v indicate earlier or later pages.
- Scrolling past the first or last real file row pages automatically.
- The menu title shows the current file name as: Seq-FILENAME
- If there are unsaved changes, it becomes: Seq-*FILENAME
- If no file is loaded, the title just says: Sequencer

Naming Mode

When naming a file or folder, the board shows this entry layout:

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z SPC - 1 2 3 4 5 <- 6 7 8 9 0 CANCEL

- Press keys to enter characters.
- SPC adds a space.
- <- deletes the previous character.
- CANCEL exits naming without saving.
- digits 0-9 are available directly on the bottom two rows.
- Press the encoder to save the name.
- the physical CANCEL key is placed lower on the board to reduce accidental exits
- naming input keys light blue at the sequencer medium light level

Status And Display

- The selected step blinks.
- The overview screen shows 8 steps at a time.
- Empty steps are dark when idle.
- If DisplayNotes is enabled, the lower playable keyboard in Sequencer mode can show Now Playing only when no step is selected.
- The performance monitor shows: AudioEng, Mem, FS, MIDI Q, Drop, and Late while the Play/Stop key is held.
- Active sequencer overlays stay awake while you are using them.
- The top title shows the loaded file name and unsaved state.